TITLE: **When is OLS BLUE? The Gauss–Markov assumptions in a regression of Bitcoin on macro-financial variables**

SUBJECT: Aim, method and results of version 2 of the `bitcoin-macro-blueml` repository, an application of undergraduate econometrics

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Key finding. Version 1 regressed the monthly change of the Bitcoin price in US dollars on changes of gold, the S&P 500, the dollar index, consumer prices and the federal funds rate, and found that OLS is not BLUE. With the same data, the failure comes from the form of the variables, not from Bitcoin: in dollar changes the error variance grows with the price level (Breusch–Pagan $p = 0.039$, White $p = 0.013$) and the residuals are autocorrelated (Breusch–Godfrey $p = 0.008$); in log returns none of the checks rejects its assumption. A random forest that explains 0.87 of the variance in sample explains none of it in new months.

1. Data and design

- **Data.** The file of version 1: 127 months, January 2015 to July 2025, Bitcoin, gold, the S&P 500 and the dollar index (Investing.com), the effective federal funds rate (FRED) and the CPI-U (FRED, BLS).[1]
- **Model.** $y_t = \beta_0 + \beta_1 g_t + \beta_2 s_t + \beta_3 d_t + \beta_4 \pi_t + \beta_5 r_t + u_t$ for Bitcoin (y), gold (g), the S&P 500 (s), the dollar index (d), consumer prices (π) and the federal funds rate (r), 126 months. **Version 1:** first differences of the levels, $\Delta x_t = x_t - x_{t-1}$. **Version 2:** log returns in per cent, $100(\ln P_t - \ln P_{t-1})$, for Bitcoin, gold, the S&P 500 and the dollar index; monthly inflation $100(\ln \mathrm{CPI}_t - \ln \mathrm{CPI}_{t-1})$; the change of the federal funds rate in percentage points.

BOX: THE GAUSS–MARKOV THEOREM

In $y = X\beta + u$, assume (A1) linearity in β, (A2) full column rank of X, (A3) $E[u \mid X] = 0$ and (A4) $\mathrm{Var}(u \mid X) = \sigma^2 I$: constant variance and no correlation across months. Then $b = (X'X)^{-1}X'y$ has $E[b] = \beta$ and $\mathrm{Var}(b) = \sigma^2(X'X)^{-1}$. For any other estimator $\tilde{b} = Cy$ that is linear in y and unbiased, $\mathrm{Var}(\tilde{b}) - \mathrm{Var}(b)$ is positive semi-definite: OLS is the *best linear unbiased estimator*. Normality of u is not needed; it makes t and F exact in small samples.

[1] As named by version 1 (for the CPI-U, FRED in its notebook, BLS in its README); the daily value that stands for a month was not recorded. The first date, 2015-01-01 04:50:10, is read as January 2015.

2. Analysis 1: the data and the form of the variables (Figure 1)

Why. The theorem concerns a linear model with constant error variance, so the form in which the variables enter decides whether A4 can hold.

Method. ADF and Phillips–Perron unit-root tests (version 1's tests) on levels, differences and returns.

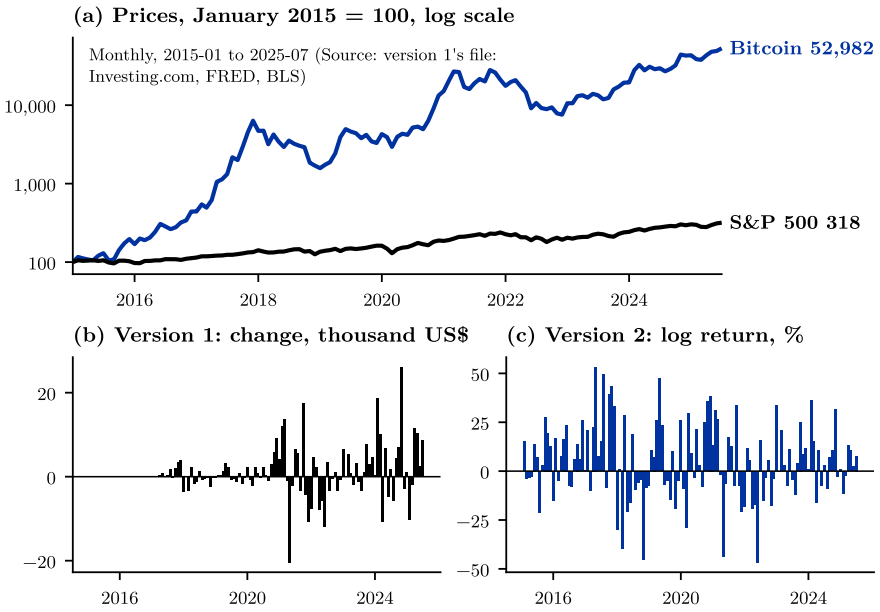


Figure 1. (a) Bitcoin and the S&P 500, January 2015 = 100, log scale. (b) Version 1's dependent variable, the monthly change of Bitcoin in thousand US dollars. (c) Version 2's dependent variable, the monthly log return in per cent.

Finding. Bitcoin rose from 218.5 to 115,765 dollars (+52,882%), the S&P 500 from 1,995 to 6,339 (+218%). ADF rejects a unit root in no level ($p \geq 0.13$) and in every log return ($p < 0.001$). The levels of Bitcoin and the S&P 500 correlate at 0.94, their returns at 0.36: two trending series correlate whether or not they are related (Granger and Newbold, 1974). The change of the CPI is ambiguous: ADF does not reject a unit root ($p = 0.37$, with 10 lags chosen by AIC for a series that is not seasonally adjusted), Phillips–Perron does ($p < 0.001$).

Interpretation. Panel (b) shows the problem before any test: a dollar change of Bitcoin in 2015 is measured in tens of dollars, in 2024 in tens of thousands. Panel (c) has the same scale throughout.

3. Analysis 2: the assumptions, one by one (Figure 2, Tables 1 and 2)

Method. A1: Ramsey's RESET adds $\hat{y}^2$ and $\hat{y}^3$ and tests them with F. A2: variance inflation factors, $\text{VIF}_j = 1/(1 - R_j^2)$. A4, constant

variance: Breusch–Pagan regresses $\hat{u}_t^2$ on the regressors, White also on their squares and products; both use $nR^2 \sim \chi^2$. A4, no autocorrelation: Breusch–Godfrey regresses $\hat{u}_t$ on the regressors and $\hat{u}_{t-1}, \ldots, \hat{u}_{t-p}$, $nR^2 \sim \chi^2_p$; Durbin–Watson $\sum(\hat{u}_t - \hat{u}_{t-1})^2 / \sum \hat{u}_t^2$ looks only at the first lag and its bounds leave an inconclusive region. A3 has no test: it concerns the unobserved u. Robust standard errors: White, $\widehat{\mathrm{Var}}(b) = \frac{n}{n-k}(X'X)^{-1}\big(\sum_t \hat{u}_t^2 x_t x_t'\big)(X'X)^{-1}$, and Newey–West with 2 lags as in version 1.

Table 1. Checks of the Gauss–Markov assumptions, 126 months. Statistic and p-value.

		Version 1: differences		Version 2: returns	
	Test	stat.	p	stat.	p
A1	RESET (F)	0.57	0.569	1.04	0.356
A2	largest VIF	1.73	–	1.67	–
A4	Breusch–Pagan	11.74	0.039	2.04	0.844
A4	White	36.70	0.013	13.87	0.837
A4	Breusch–Godfrey, 2 lags	9.60	0.008	1.73	0.421
A4	Breusch–Godfrey, 6 lags	17.14	0.009	1.76	0.940
A4	Durbin–Watson	1.67	–	1.78	–
–	Jarque–Bera (normality)	107.92	<0.001	5.07	0.079

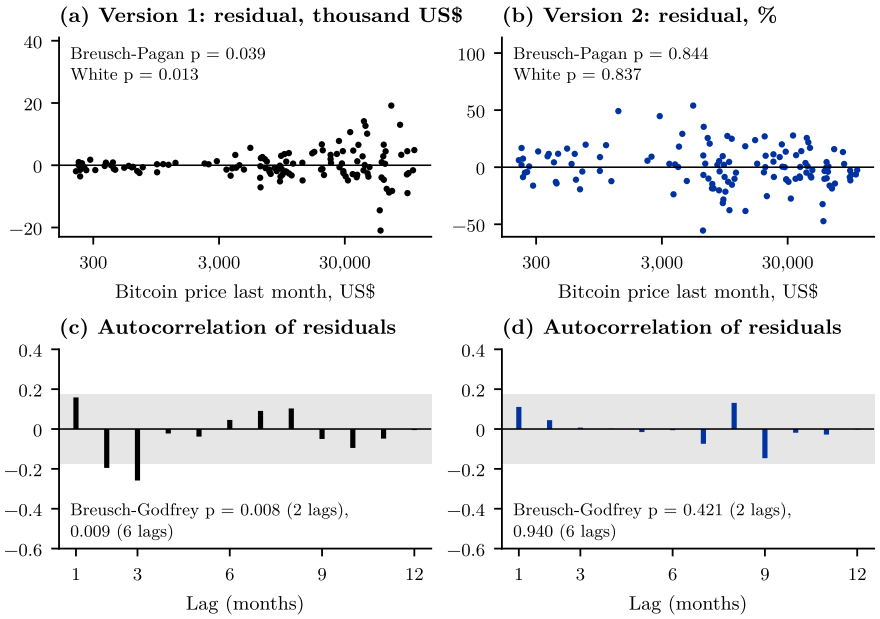


Figure 2. Residuals of the two regressions. (a), (b) Against the Bitcoin price of the previous month (log scale). (c), (d) Autocorrelations; grey band $\pm 1.96/\sqrt{n}$.

Finding. In version 1's form, A4 fails twice: the error variance depends on the regressors (Breusch–Pagan $p = 0.039$, White $p = 0.013$), and the absolute residual rises with last month's Bitcoin price (rank correlation 0.57, p <0.001); the residuals are autocorrelated (Breusch–Godfrey $p =$ 0.008 with two lags, 0.009 with six). In version 2's form neither part of A4 is rejected (Breusch–Pagan $p = 0.844$, White 0.837, Breusch–Godfrey 0.421 and 0.940), nor A1 (RESET $p = 0.356$); the rank correlation with the price is -0.08 ($p = 0.351$). Collinearity is no issue in either form (largest VIF 1.73 and 1.67). Normality is rejected for version 1 (Jarque–Bera p <0.001) but not for version 2 ($p = 0.079$).

Table 2. Version 2: coefficients with classical, White (HC1) and Newey–West (2 lags) standard errors; p-value with White standard errors. $R^2 = 0.155$.

	coef.	classical	White	Newey–West	p
constant	4.77	2.12	2.24	2.48	0.033
gold	0.49	0.50	0.45	0.40	0.280
S&P 500	1.67	0.39	0.36	0.33	<0.001
dollar index	0.93	1.09	0.96	0.93	0.331
inflation	−5.68	5.12	5.36	5.80	0.289
fed funds rate (change)	−7.94	9.47	8.98	9.13	0.377

Interpretation. With A1, A2 and A4 not rejected, OLS on returns is BLUE provided A3 holds. Only the S&P 500 matters: a month in which its log return is 1 point higher goes with a Bitcoin log return 1.67 points higher (White s.e. 0.36); the other variables have $p \geq 0.27$. The three standard errors differ little, as expected when A4 holds.

4. Analysis 3: what BLUE means, in a simulation (Figure 3)

Method. The regressors of version 2 are held fixed, β and σ are set to their estimates, and 20,000 samples $y = X\beta + u$ are drawn under (A) $u_t \sim N(0, \sigma^2)$ and (B) $\mathrm{Var}(u_t) \propto 0.25 + z_t^2$, z_t the standardised S&P 500 return, with the same average variance. Two linear unbiased estimators of β_2 are compared: OLS and weighted least squares, $b_W = (X'WX)^{-1}X'Wy$ with $W = \mathrm{diag}\{1/(0.25 + z_t^2)\}$, which is generalised least squares under B.

Finding. Both estimators are unbiased under both laws (means 1.664 and 1.665 under A, 1.666 and 1.669 under B). Under A, OLS has the smaller standard deviation (0.39 against 0.57), as the theorem says. Under B, WLS does (0.47 against 0.68), and the classical 95% interval of OLS contains the true value in only 72.9% of samples; White's interval in 90.6%.

Interpretation. A failure of A4 does not bias OLS; it makes it inefficient and its classical standard errors wrong. White's standard errors repair most, not all, of the coverage in a sample of this size.

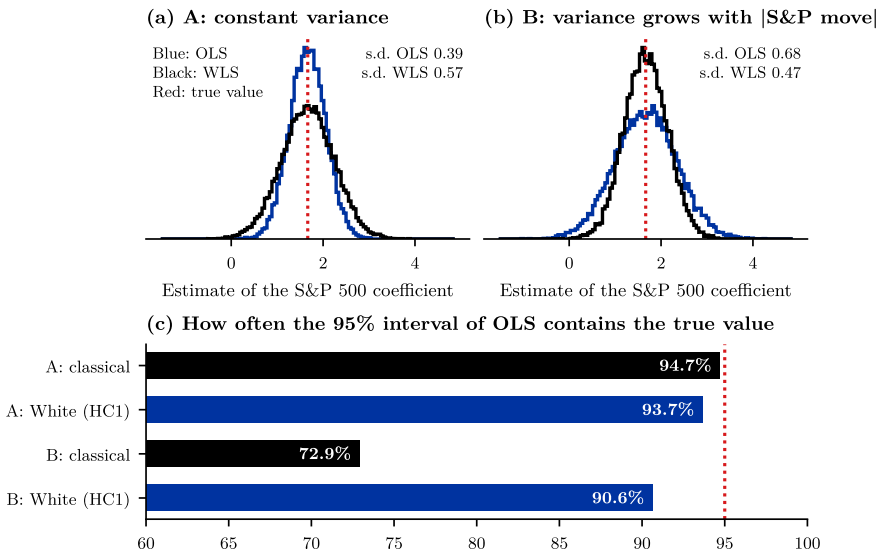


Figure 3. (a), (b) Distribution of the OLS and WLS estimates of the S&P 500 coefficient under A and B; red: true value 1.67. (c) Share of samples in which the 95% interval of OLS contains the true value.

5. Analysis 4: the random forest (Figure 4)

Method. Version 1's forest (500 trees, fully grown, seed 42) and OLS, first on all months, then for each of the last 60 months (2020-08 to 2025-07) fitted on the earlier months only and applied to that month: $R^2_{\text{new}} = 1 - \sum(y_t - \hat{y}_t)^2 / \sum(y_t - \bar{y}_t)^2$, $\bar{y}_t$ the mean of the earlier months.

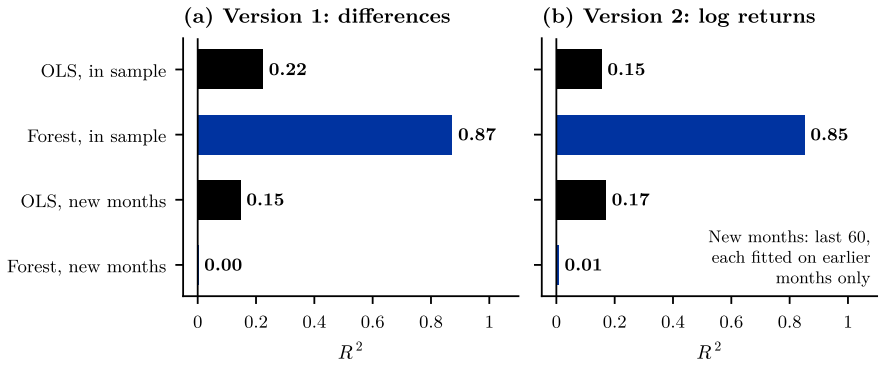


Figure 4. R^2 of OLS and of the random forest in sample and on months not used in fitting.

Finding. In sample the forest reaches $R^2 = 0.87$ (version 1's form) and 0.85 (returns), against 0.22 and 0.15 for OLS. On new months the forest falls to 0.00 and 0.01, OLS keeps 0.15 and 0.17.

Interpretation. With 126 months, fully grown trees memorise the sample;

the in-sample R^2 of version 1 measured that. The linear model carries over to new months; the forest finds no additional structure.

6. Conclusions and limitations

1. Whether OLS is BLUE depends on the form of the variables: in dollar changes A4 fails, in log returns it is not rejected.
2. Only the S&P 500 is related to Bitcoin in the same month; gold, the dollar, inflation and the policy rate are not.
3. A3 cannot be tested. Bitcoin and the S&P 500 react to the same news in the same month, so the coefficient describes co-movement, not the effect of one on the other; version 1's Granger tests find no lead of the S&P 500 over Bitcoin ($p \geq 0.19$ for 1–6 lags).

Scope and verification note. Every number is written from `figures/*.csv` by `make_paper.py`; the unit tests check the computations against matrix formulas, auxiliary regressions written out, exact variances for the simulation and the printed output of version 1, whose code reproduces it on the stored file. The sample is 126 months; the monthly timing of the source series was not recorded.

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An AI coding assistant was used in writing the code and in preparing this note.